

HealthGuard ML Report

This automated report summarizes the machine learning training outputs for HealthGuard.

Included sections: dataset summary, model comparison, confusion matrices, ROC curves, clustering results, feature importance analysis and best model summary.

Dataset Summary

Metric	Value
Raw rows	1025.0000
Rows after duplicate removal	302.0000
Columns	14.0000
Feature columns	13.0000
Missing values	0.0000
Duplicate rows removed	723.0000
Heart disease cases	164.0000
No heart disease cases	138.0000

Dataset Notes

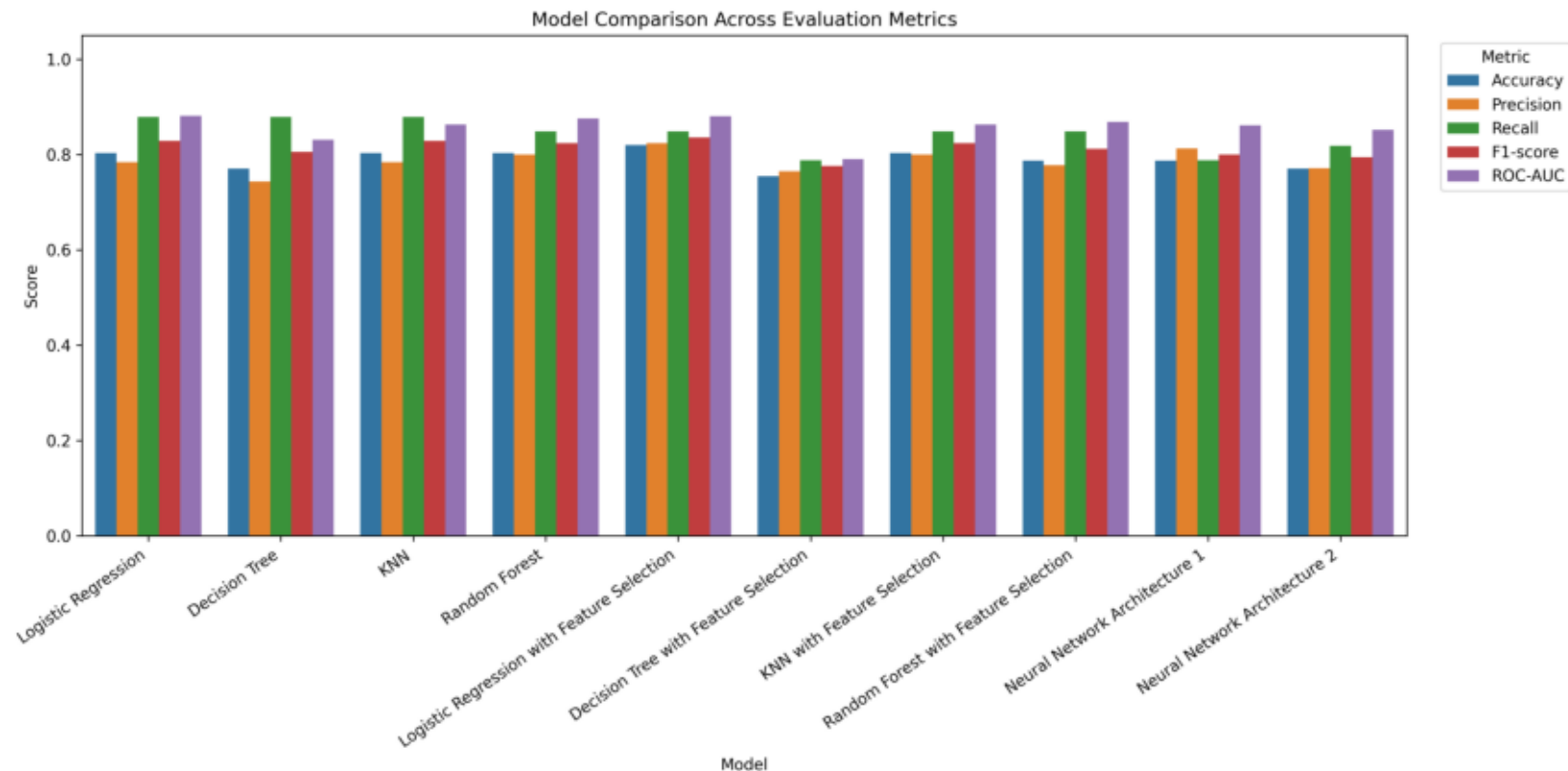
The report is generated from the Heart Disease dataset used by the HealthGuard machine learning notebook. Duplicate rows are removed in the notebook before model training, so the cleaned dataset counts are shown here.

The target variable uses 1 for heart disease and 0 for no heart disease. All model, clustering and feature importance summaries in this PDF are generated from saved training outputs in ml/results.

Model Comparison Table

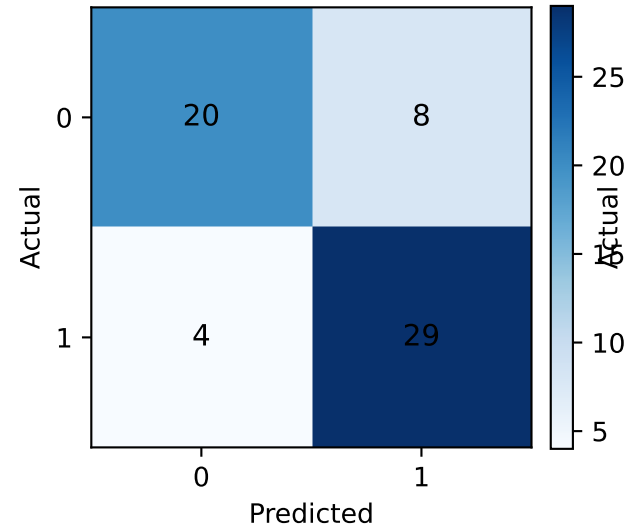
Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.8033	0.7838	0.8788	0.8286	0.8810
Decision Tree	0.7705	0.7436	0.8788	0.8056	0.8306
KNN	0.8033	0.7838	0.8788	0.8286	0.8626
Random Forest	0.8033	0.8000	0.8485	0.8235	0.8755
Logistic Regression with Feature Selection	0.8197	0.8235	0.8485	0.8358	0.8799
Decision Tree with Feature Selection	0.7541	0.7647	0.7879	0.7761	0.7900
KNN with Feature Selection	0.8033	0.8000	0.8485	0.8235	0.8626
Random Forest with Feature Selection	0.7869	0.7778	0.8485	0.8116	0.8680
Neural Network Architecture 1	0.7869	0.8125	0.7879	0.8000	0.8615
Neural Network Architecture 2	0.7705	0.7714	0.8182	0.7941	0.8517

Model Comparison Chart

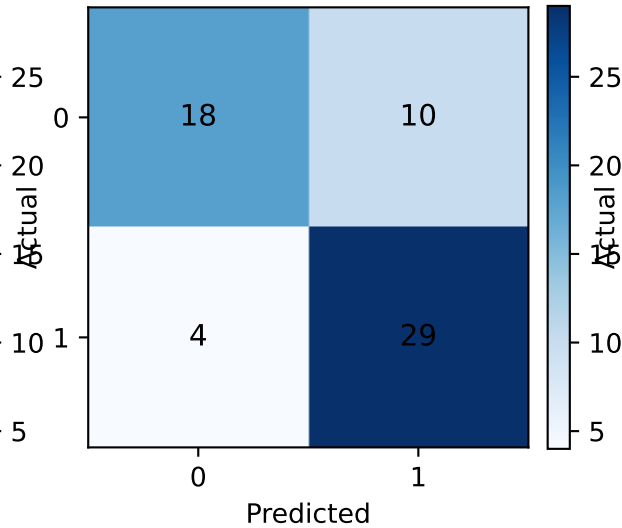


Confusion Matrices

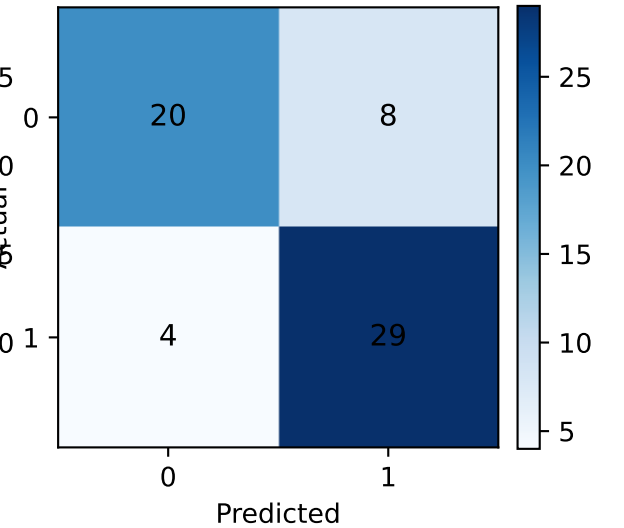
Logistic Regression



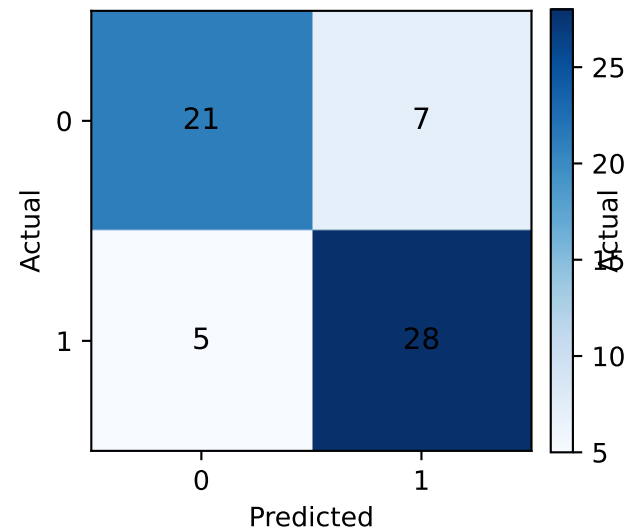
Decision Tree



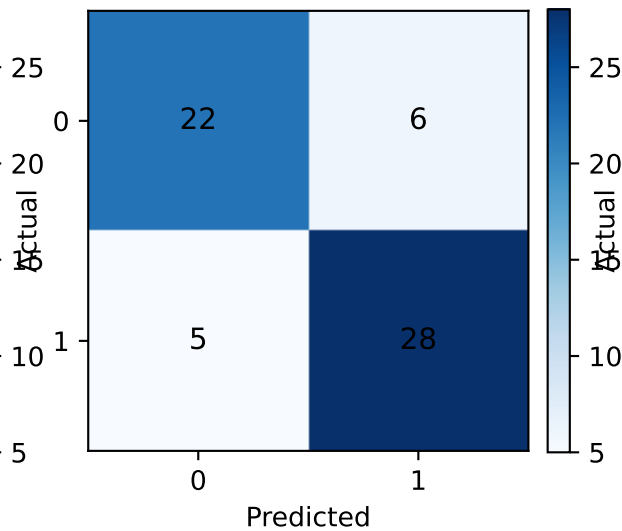
KNN



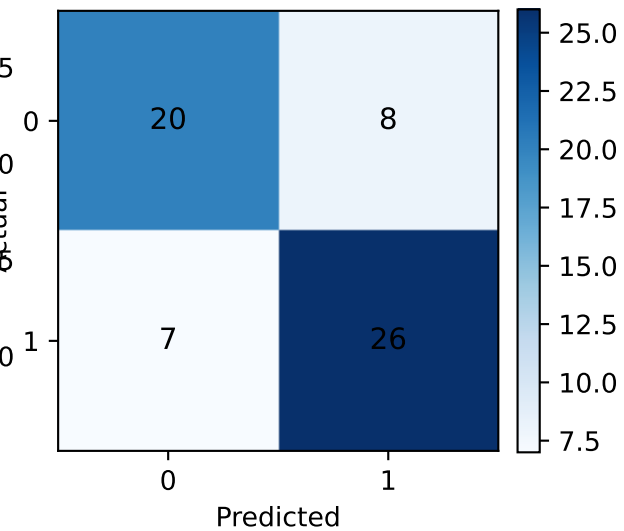
Random Forest



Logistic Regression with Feature Selection

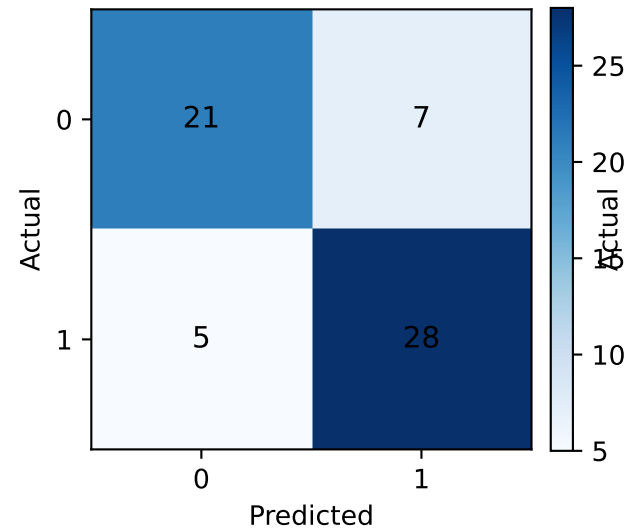


Decision Tree with Feature Selection

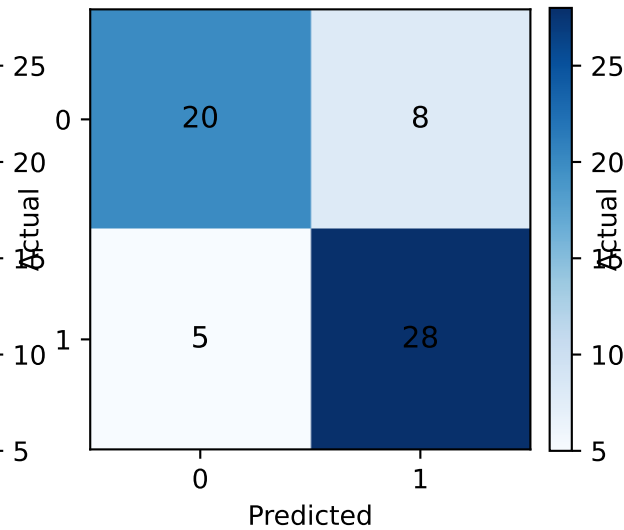


Confusion Matrices

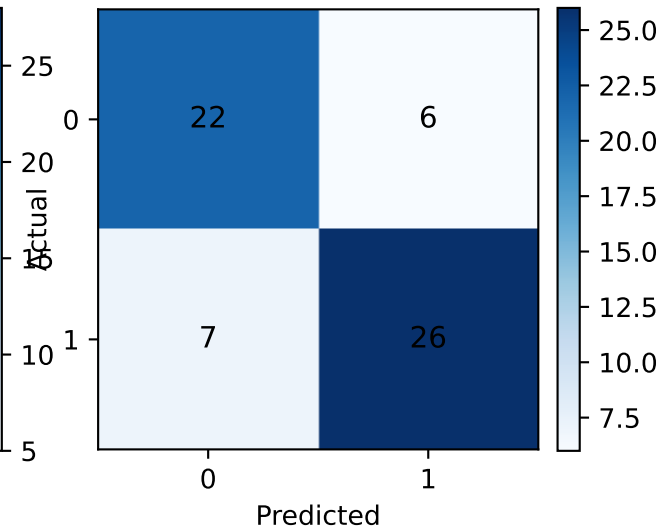
KNN with Feature Selection



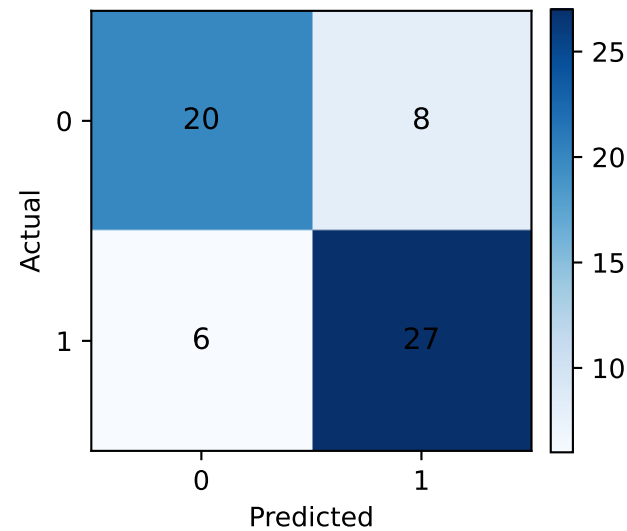
Random Forest with Feature Selection



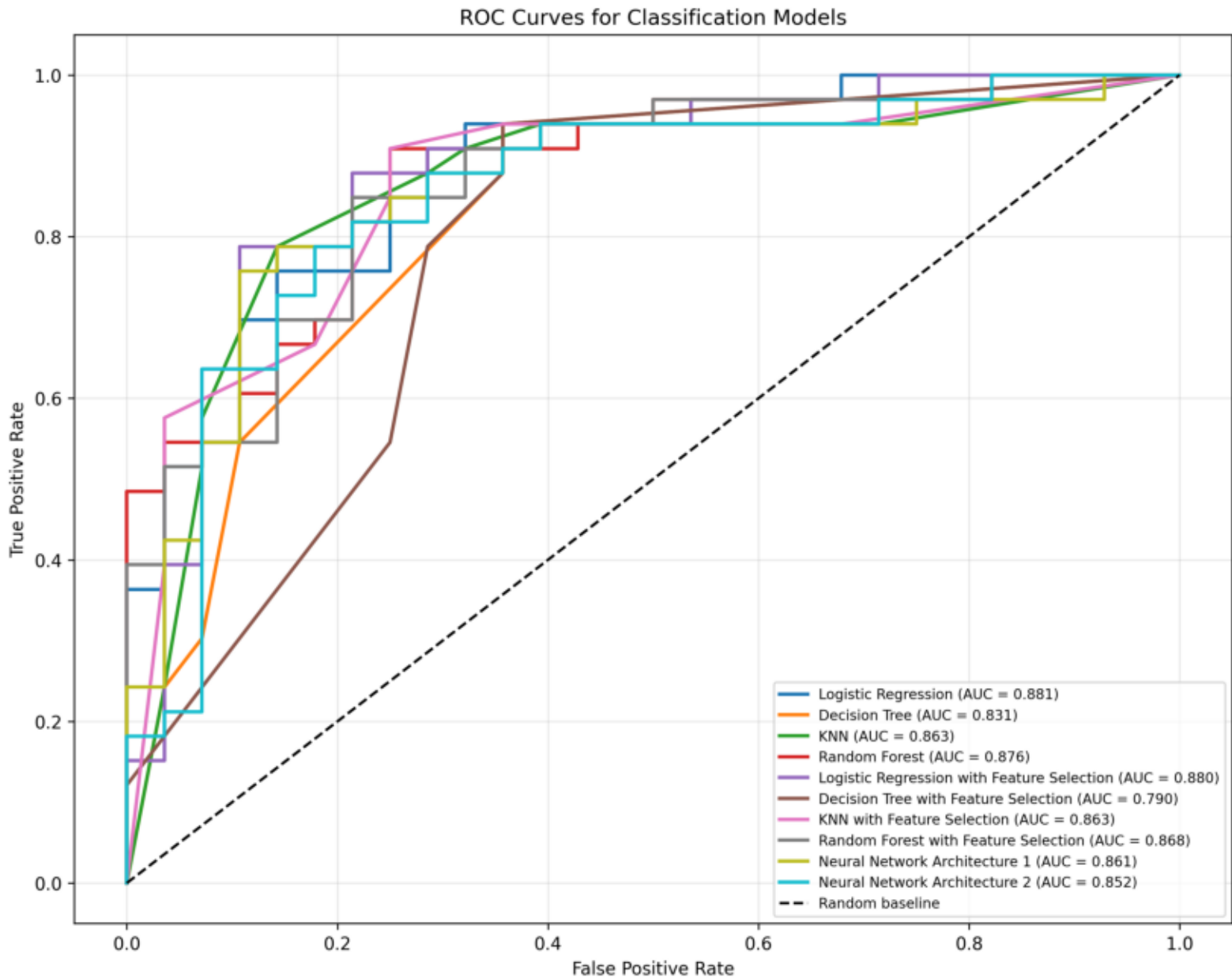
Neural Network Architecture 1



Neural Network Architecture 2



ROC Curves



Best Model Summary

Best model by F1-score: Logistic Regression with Feature Selection.

Key metrics: Accuracy=0.8197, Precision=0.8235, Recall=0.8485, F1-score=0.8358, ROC-AUC=0.8799.

F1-score is used as the primary summary metric because it balances precision and recall, which is important when heart disease screening must manage both false positives and false negatives.

Cross-Validation Stability

	Model	Accuracy Mean	Accuracy Std	Precision Mean	Precision Std	Recall Mean	Recall Std	F1 Score Mean	F1 Score Std
KNN with Feature Selection	KNN with Feature Selection	0.8543	0.0223	0.8422	0.0505	0.9087	0.0505	0.8716	0.0162
	Random Forest with Feature Selection	0.8479	0.0391	0.8307	0.0529	0.9146	0.0887	0.8661	0.0371
Logistic Regression with Feature Selection	Logistic Regression with Feature Selection	0.8477	0.0321	0.8362	0.0438	0.9025	0.0799	0.8644	0.0319
	Logistic Regression	0.8279	0.0450	0.7953	0.0527	0.9267	0.0626	0.8539	0.0390
Neural Network Architecture A	Neural Network Architecture A	0.8378	0.0432	0.8404	0.0424	0.8718	0.1024	0.8512	0.0484
	Random Forest	0.8279	0.0368	0.8090	0.0496	0.9023	0.0678	0.8502	0.0325
Neural Network Architecture B	Neural Network Architecture B	0.8213	0.0336	0.8347	0.0577	0.8475	0.0836	0.8361	0.0352
	KNN	0.8046	0.0263	0.7937	0.0198	0.8657	0.0597	0.8270	0.0286
Decision Tree with Feature Selection	Decision Tree with Feature Selection	0.7713	0.0635	0.7487	0.0587	0.8845	0.0749	0.8082	0.0468
	Decision Tree	0.7614	0.0615	0.7449	0.0558	0.8663	0.0908	0.7973	0.0483

Model Stability Notes

Cross-validation reports mean and standard deviation across five stratified folds. Higher mean values indicate stronger average performance; lower standard deviations indicate more stable behavior across different patient subsets.

Models with similar F1-score means should be compared by their F1-score standard deviation. A lower standard deviation suggests the model is less dependent on one favorable train-test split.

Clustering Results

k	Inertia	Silhouette Score	Adjusted Rand Index	Normalized Mutual Information
2.0000	2652.9010	0.1683	0.4110	0.3502
3.0000	2438.2792	0.1171	0.2558	0.2595
4.0000	2266.0572	0.1209	0.1826	0.1902
5.0000	2153.9783	0.1187	0.1626	0.1882
6.0000	2065.4597	0.1120	0.1299	0.1879
7.0000	2000.6129	0.1261	0.1500	0.1943
8.0000	1935.8368	0.1100	0.1118	0.1905
9.0000	1877.0214	0.1041	0.1206	0.2021
10.0000	1811.7163	0.0998	0.1188	0.2114

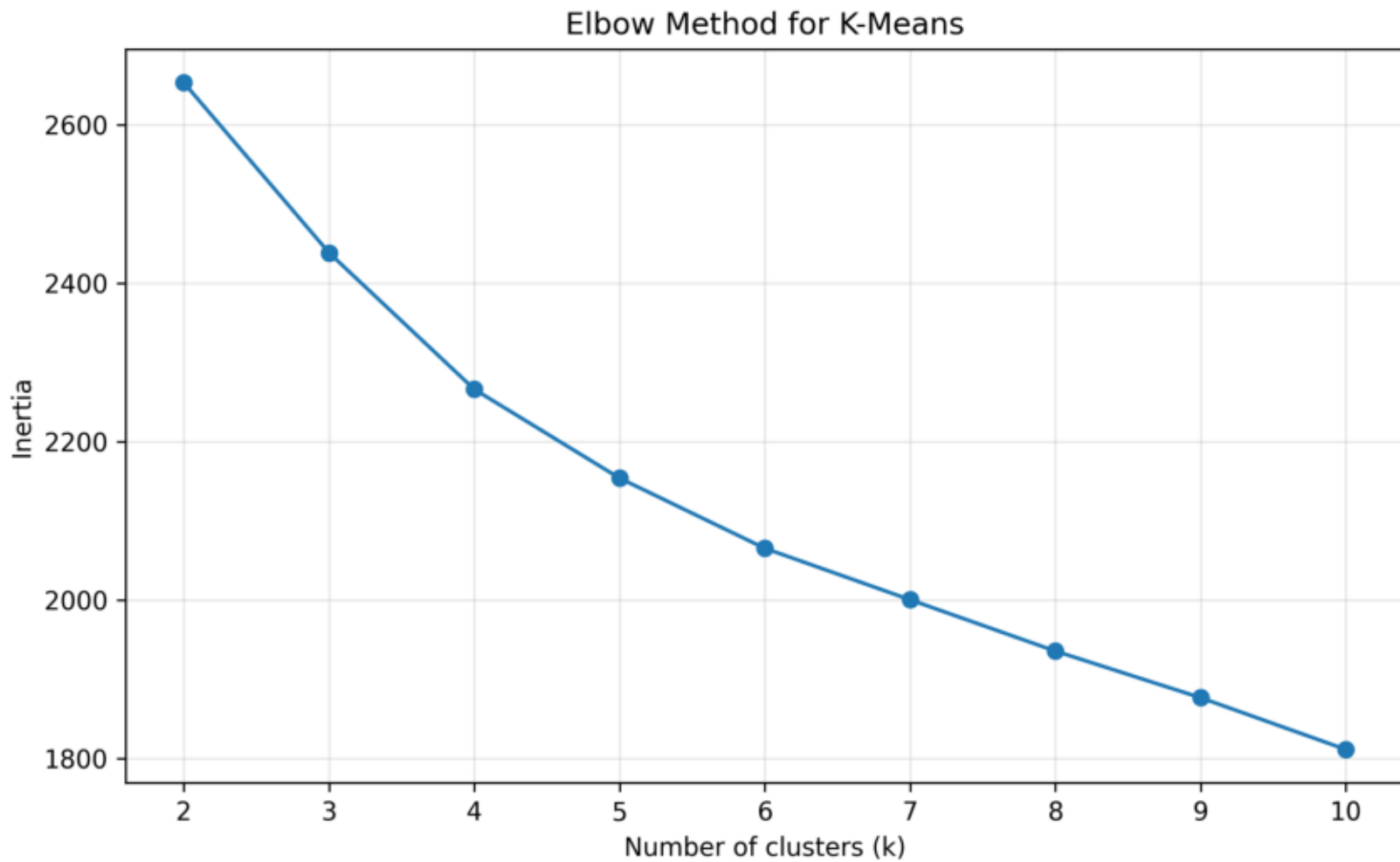
Clustering Summary

The best cluster count selected by Silhouette Score was $k=2$.

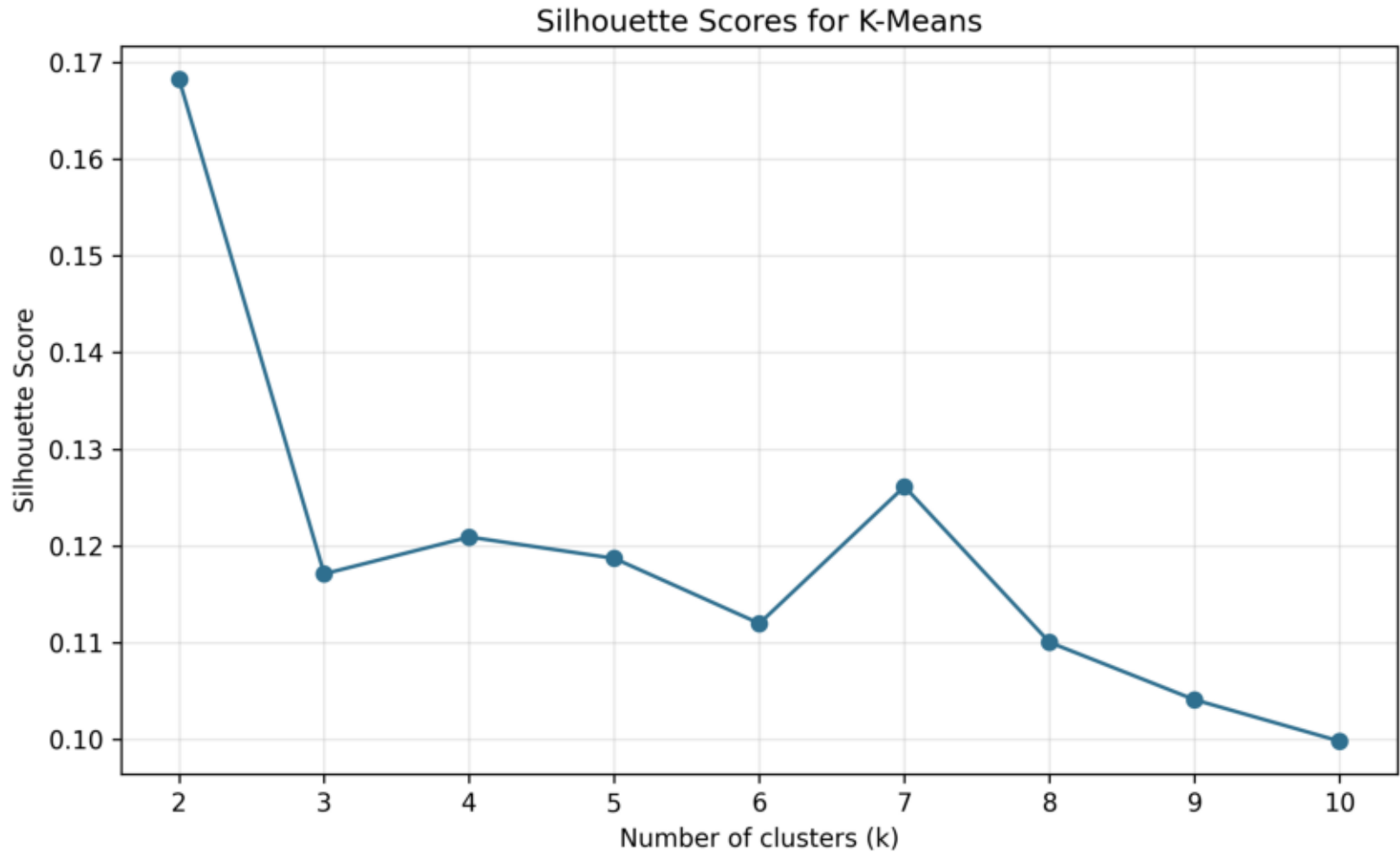
For this k , Silhouette Score=0.1683, Adjusted Rand Index=0.4110, and Normalized Mutual Information=0.3502.

ARI and NMI compare unsupervised cluster assignments with the known heart disease labels. These metrics are diagnostic only; the labels are not used by K-Means during clustering.

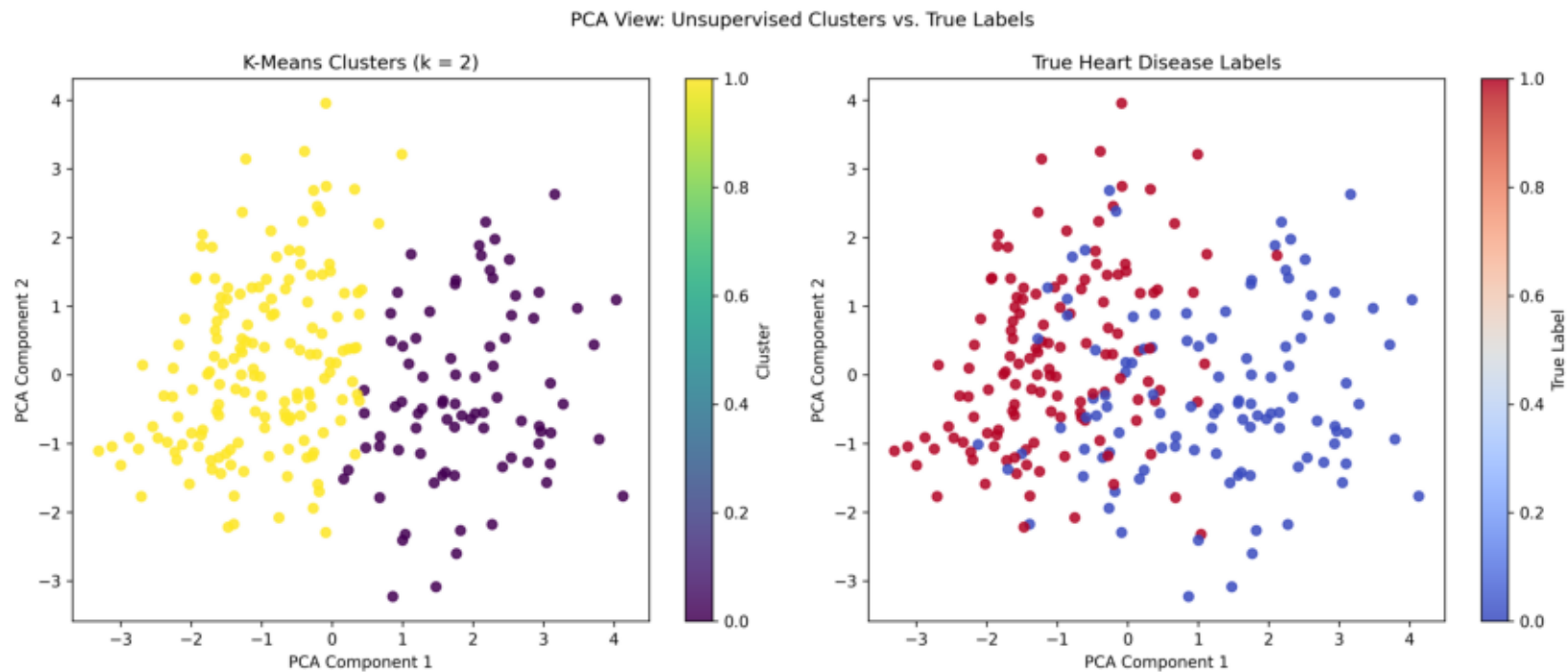
Elbow Method



Silhouette Scores



PCA Cluster Visualization

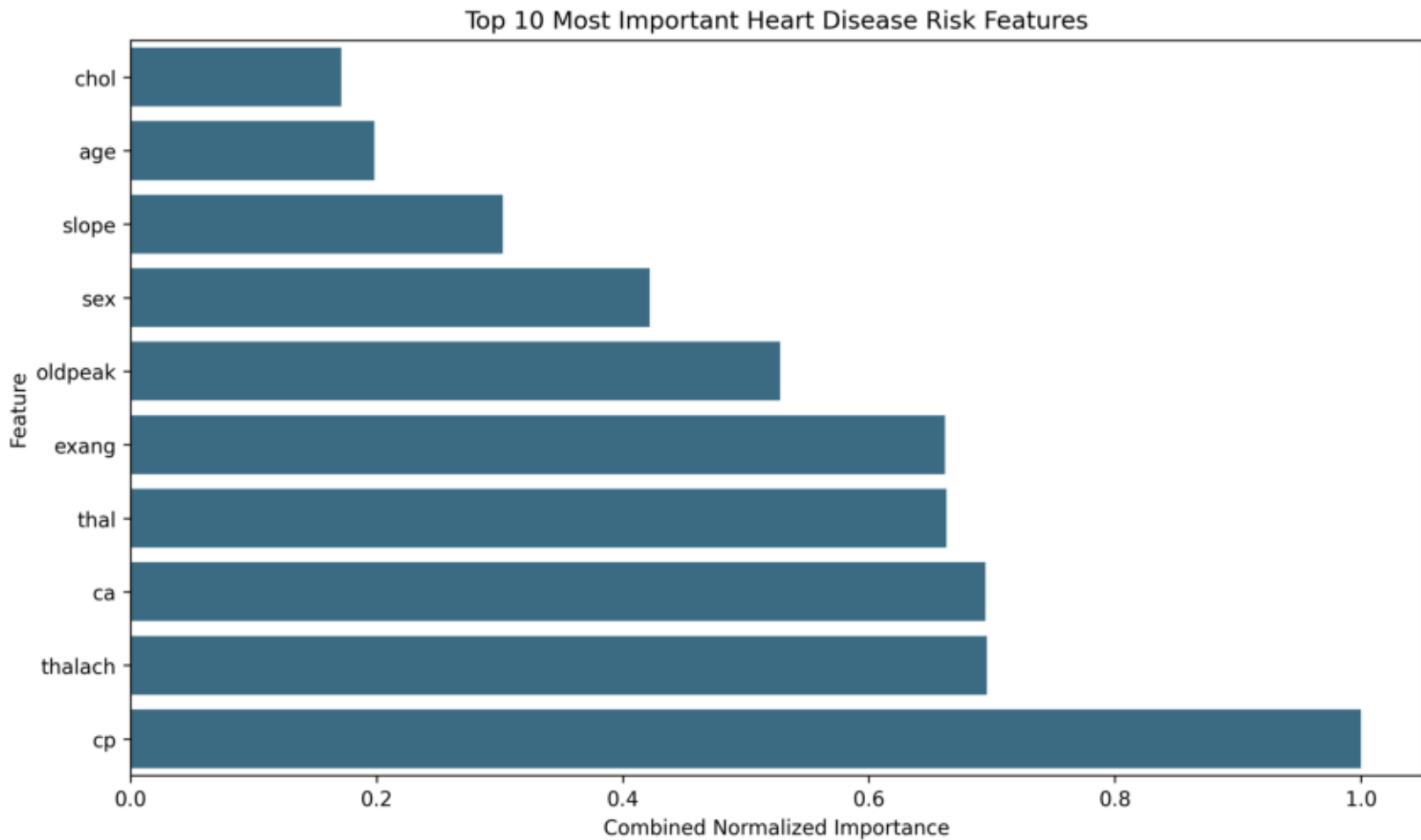


The left plot shows K-Means cluster assignments. The right plot shows the true heart disease labels for the same PCA-projected patients.

Top 10 Feature Importance

Overall Rank	Feature	Logistic Regression Coefficient	Random Forest Importance	Combined Importance
1.0000	cp	0.2845	0.2102	1.0000
2.0000	thalach	0.2113	0.1364	0.6958
3.0000	ca	-0.2142	0.1338	0.6947
4.0000	thal	-0.1907	0.1379	0.6631
5.0000	exang	-0.2248	0.1121	0.6618
6.0000	oldpeak	-0.1812	0.0880	0.5277
7.0000	sex	-0.1884	0.0381	0.4218
8.0000	slope	0.1269	0.0335	0.3027
9.0000	age	-0.0559	0.0421	0.1984
10.0000	chol	-0.0547	0.0315	0.1712

Feature Importance Chart



Feature Importance Analysis

The leading features in the combined importance ranking were: cp, thalach, ca, thal, exang.

Logistic Regression coefficients provide direction: positive coefficients increase the predicted risk score and negative coefficients decrease it. Random Forest importances rank how strongly each feature contributes to tree split quality, but they do not provide a positive or negative direction.